

LOUISA B. EBBY

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RESEARCH INTERESTS

Data-driven modeling and state estimation techniques with applications to environmental systems.

EDUCATION

North Carolina State University

2023 -

- Ph.D, Biomathematics
- Advised by Dr. Mohammad Farazmand
- Expected Spring 2028

Williams College

2016-2020

- B.A. *cum laude*, Mathematics & Environmental Studies

RESEARCH EXPERIENCE

Graduate Research Assistant in the Farazmand Lab

2023-

North Carolina State University

Raleigh, NC

- Collaborating with Dr. Mohammad Farazmand on data-driven modeling with sparse measurements for environmental systems.
- Developing state estimation methods from sparse measurements for dynamical systems subject to physical constraints.
- Leveraging emergency call data to improve wildfire forecasts in the wildland-urban interface.

NSF Research Experience for Undergraduates (REU) Graduate Student Mentor

Summer 2025

North Carolina State University

Raleigh, NC

- Supervised four undergraduate students on estimating sea surface temperature from partial observations.
- Guided students in developing and implementing tools in MATLAB.
- Provided mentorship in scientific reasoning, research communication, and collaborative writing; currently co-authoring a manuscript based on project results.

TEACHING EXPERIENCE

Graduate Teaching Assistant

2023-2025

North Carolina State University

Raleigh, NC

- Teaching Assistant for Calculus I, Calculus III, led recitation sessions twice per week (Spring 2024, Fall 2025).
- Teaching Assistant for Differential Equations (Fall 2023).
- Drop-in tutor for undergraduates in Math Tutoring Center (Fall 2023).

Secondary School Mathematics Teacher

2020-2023

Groton School

Groton, MA

- Courses: AP BC Calculus, Multivariable Calculus, Multivariable Calculus and Linear Algebra, Algebra 2 Honors, Summer Algebra I Review Course.
- Supervised student in independent study on differential equations.

Undergraduate Teaching Assistant*Williams College*

2019-2020

Williamstown, MA

- Courses: Multivariable Calculus, Differential Equations.
- Graded homework assignments and ran weekly help sessions.

Freelance Mathematics Tutor

2019-

- Tutor high school and undergraduate students in a variety of math courses ranging from Precalculus to Linear Algebra.

FELLOWSHIPS, HONORS & AWARDS

Graduate Research Fellowship Honorable Mention, 2025, *National Science Foundation*

Student Leadership Award, 2025, College of Sciences, *North Carolina State University*

Award of Excellence, Poster Presentation, 2025, *Society of Mathematical Biology*

NSF Research Training Group Fellow, Summer 2025/Spring 2026, *North Carolina State University*

Lucas First Year Student Award, 2024, Biomathematics Graduate Program, *North Carolina State University*

Franke-Norris-Griggs Award for Service, 2026, Mathematics Department *North Carolina State University*

University Graduate Fellowship, 2023-2024, *North Carolina State University*

SERVICE & OUTREACH

Chapter Officer

2023-

*Association for Women in Mathematics (AWM)**NC State*

- Organized Sonia Kovalevsky (SK) Day, an outreach event with workshops to inspire an interest in mathematics and problem-solving for local students in grades 6-12 (Spring 2025, 2026).
- Planned and ran an applied mathematics workshop during SK Day (Spring 2024).
- Chapter Treasurer (Summer 2024-Spring 2026).
- Chapter President (Spring 2026-present)
- Organize Women in Math monthly lunches for the math community (Fall 2024-).

Co-Organizer

2023-

*Triangle Competition in Math Modeling (TriCoMM)**NC State*

- Organize NC State portion of joint local modeling competition between NC State, UNC Chapel Hill and Duke University for undergraduate students.
- Collaborate with organizers at other institutions to write competition problems.
- Evaluate and give feedback on competition entries.

Student Seminar Chair

2024-2025

*Biomathematics Seminar, Biomathematics Graduate Student Association**NC State*

- Collaborated with faculty to invite and coordinated speakers for weekly seminar.

Mentor

2023-2025

*Undergrads Union Grads (UUG)**NC State*

- Mentored undergraduate mathematics students in future career goals.

Graduate Program Representative

Biomathematics Graduate Program, 2026 Joint Mathematics Meeting, Washington D.C.

January 2026

NC State

- Represented program at graduate program recruitment fair.

PUBLICATIONS & PREPRINTS

L.B. Ebby, M. Farazmand. *Harnessing eyewitness reports improves real-time forecasts of fire spread in the WUI*. Submitted June 2026.

L. B. Ebby, M. Farazmand. *Discrete Empirical Interpolation Method with upper and lower bound constraints*. Journal of Computational Science, vol. 100, pp. 102983, 2026.

C. All, K. Ho, M. Magnuski, C. Nicolaides, **L. B. Ebby**, M. Farazmand. *Rapid estimation of global sea surface temperatures from sparse streaming in situ observations*. Journal of Geophysical Research: Machine Learning and Computation, vol. 3 pp. e2026JH001279, 2026.

PRESENTATIONS

Society of Industrial and Applied Mathematics Conference on Uncertainty Quantification	March 2026
<i>Wildfire Perimeter Estimation from Emergency Call Data</i>	Minneapolis, MN

NCSU NSF Research Training Group Seminar	February 2026
<i>Estimation of Global Sea Surface Temperatures from Sparse in situ Observations</i>	Raleigh, NC

American Physical Society Division of Fluid Dynamics Annual Meeting	November 2025
<i>Flow Reconstruction from Sparse Measurements with Upper and Lower Bound Constraints</i>	Houston, TX

Data Meets Dynamics Workshop (Lightning Talk + Poster)	August 2025
<i>Wildfire Prediction from Sparse Observational Measurements</i>	Madison, WI

Society for Mathematical Biology Annual Meeting (Poster)	July 2025
<i>Wildfire Prediction from Sparse Observational Measurements</i>	Edmonton, AB, Canada

- Received Poster Award of Excellence

NC State Graduate School Research Symposium (Poster)	April 2025
<i>Wildfire Prediction from Sparse Observational Measurements</i>	Raleigh, NC

Triangle Area Graduate Mathematics Conference	February 2025
<i>Wildfire Prediction from Sparse Observational Measurements</i>	Durham, NC

Biomathematics Graduate Program Recruitment (Poster)	March 2025
<i>Wildfire Prediction from Sparse Observational Measurements</i>	Raleigh, NC

NCSU NSF Research Training Group Seminar	November 2024
<i>Wildfire Prediction from Sparse Observational Measurements</i>	Raleigh, NC

Triangle Computational and Applied Mathematics Symposium (Lightning Talk)	October 2024
<i>Wildfire Prediction from Sparse Observational Measurements</i>	Chapel Hill, NC

SKILLS

Mathematica, MatLab, Python, R

PROFESSIONAL AFFILIATIONS

American Mathematical Society, American Physical Society, Association for Women in Mathematics, Society for Industrial and Applied Mathematics